

NAJATH AKRAM, PH.D.

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SUMMARY

Signal processing engineer, 10+ years in wireless, the last six on LTE and 5G NR O-RAN radio units. DFE and low-PHY from architecture to bit-accurate model: DUC/DDC, CFR, channel filters, PRACH detection, BFP, fixed-point modeling. Develop architecture and software/firmware interface specifications at Airspan and previously at Jabil; owns the bit-accurate models and 7.2x fronthaul tooling for firmware sign-off. 10+ IEEE publications.

CORE SKILLS

Wireless systems: 5G NR and LTE physical layer; O-RAN fronthaul (split 7.2x, WG4 CUS-Plane, eCPRI, BFP compression); 3GPP TS 36.141, 38.104, 38.141, and 38.211; multiband TDD/FDD O-RU architecture; EVM, ACLR, OBUE, and PIM analysis

DSP and modeling: DFE design (DUC and DDC, NCO, channel filters, interpolation and decimation chains); CFR and cancellation-pulse design; PRACH detection; bit-accurate fixed-point modeling against vendor IP C-models; BFP codecs

Platforms and tools: MATLAB (5G, LTE, and HDL toolboxes, App Designer, Compiler); AMD (Xilinx) RFSoc and DFE IP; Python; Keysight 89600 VSA

AI and SW engineering: Anthropic Claude API and Agent SDK; multi-agent pipelines with schema-validated outputs; per-call cost ledgers and budgets; evaluation harnesses; TypeScript, Next.js, PostgreSQL

EXPERIENCE

AIRSPAN NETWORKS

Signal Processing Engineer

Houston, Texas

January 2025 to present

O-RAN 7.2x radio units: bit-accurate DFE and low-PHY reference modeling, FPGA IP verification, and fronthaul test tooling.

- Write the DFE architecture and the software/firmware interface specifications that the digital design and firmware teams implement, continuing the specification work from Jabil.
- Develop and maintain the bit-accurate downlink/uplink DFE and low-PHY reference models that generate the RTL golden vectors: NR and LTE test models through IFFT/FFT, CFR, the Xilinx channel-filter, mixer IP C-models, and O-RAN BFP at 8, 9, 10, 12, and 14 bits, across 3 to 100 MHz including combined-band carriers (e.g., B28+B20).
- Built the bit-accurate PRACH receive-chain model (Xilinx PRACH DDC and decimation, CP removal, FFT, subcarrier extraction, BFP and U-plane packing, detection) covering every LTE format and NR FR1 formats, FDD and TDD.
- Built an O-RAN fronthaul power and PAPR tool: per-eAxC dBFS power, grid-domain PAPR, per-tone PAPR, and oversampled time-domain PAPR reported separately, auto-detecting BFP width and eAxC bit split and deriving the compressed-domain full-scale reference per O-RAN WG4 CUS-Plane section 8.1.3.1. Shipped as MATLAB Compiler GUI and CLI executables plus a Python version.
- Built an uplink EVM and PRACH analyzer that demodulates PUSCH and PRACH from 7.2x U-plane captures, with views aligned to the Keysight 89600 VSA, and an RU-to-DU full-scale alignment check that emulates the DU's fixed-point ingest against the per-PRB BFP exponents to separate RU misconfiguration from DU ingest errors in FS offset.
- Built an expected versus captured IQ comparison GUI used in bring-up: cross-correlation delay alignment, overlaid PSD and PSD difference, and dBFS power readout for hex, BFP, .mat, and raw binary captures.
- Re-clocked CFR from 491.52 to 245.76 MSPS and redesigned the cancellation pulses and channel filters for 3 to 20 MHz to meet an operator's 3 MHz carrier requirement, tuning cutoff and pulse length against a per-symbol constellation EVM metric rather than the vendor's RMS figure: 2.17 to 2.23% EVM at inside ACLR and OBUE margin.
- Built a multiband PIM planner that catalogs IM3, IM5, and IM7 products for multi-carrier downlink and flags uplink-band hits; checked by hand against a B71, B29, and B14 stack.

JABIL INC.

Principal Wireless Systems Design Engineer

Lead Wireless Systems Design Engineer

Senior FPGA Design Engineer

Warren, New Jersey and Houston, Texas

December 2023 to August 2024

September 2021 to December 2023

December 2020 to September 2021

- Led the firmware systems workstreams for LTE and 5G NR O-RAN radio units from single-band to tri-band through commercial release: defined NR implementation requirements, wrote the architecture and implementation

documents the digital design team built to, weighed massive MIMO and O-RAN functional-split options against implementation complexity, resources, and timelines, and briefed customers on passive intermodulation.

- Primary author of the DFE System Design Documents for two radio lines, a B3 FDD radio (5 to 40 MHz) and a 100 MHz TDD radio: the interpolation chain to 983.04 MSPS, pulse-shaping filters derived from TS 38.104 guard-band rules and grouped into five classes covering 13 bandwidths, CFR cancellation-pulse configuration, and uplink decimation and PRACH routing.
- Authored the firmware implementation guides for the NR symbol phase-compensation term of TS 38.211 section 5.4 and for RFSoc ZU67DR pre-emphasis (per-subcarrier 16-bit LUTs flattening pulse-shaping passband droop for 5 to 30 MHz, with the LTE and LTE + NB-IoT edge cases per TS 38.104 section 5.7.3).
- Owned filter design, CFR simulation, and cancellation-pulse generation for uplink, downlink, and PRACH DFE firmware.
- Generated RTL stimulus vectors, integrated Xilinx hard-IP C-models, developed executable models for UVM testbenches, and wrote PCAP tooling to extract O-RAN C-plane and U-plane data for bit-accurate system modeling.

FLORIDA INTERNATIONAL UNIVERSITY

Miami, Florida

Graduate Research Assistant

2018 to 2020

- Developed digital and mixed-domain hardware-reduction algorithms for massive MIMO receivers, the basis of the dissertation and most of the 10+ IEEE publications, all on reduced-ADC array architectures.
- Implemented a 28 GHz digital beamforming array receiver with 800 MHz per channel on Xilinx RFSoc (ZCU111, ZCU1275, ZCU1285), cutting ADC count by 75% with frequency multiplexing, and showing a further 50% ADC reduction for 2D phased arrays over the air.

MATHWORKS

Natick, Massachusetts

Engineering Intern

2019

- Enhanced HDL Verifier workflows and developed Ethernet PHY-interface IP across Artix, Kintex, and Virtex FPGAs; two awards in the fall 2019 intern hackathon.

THE UNIVERSITY OF AKRON

Akron, Ohio

Graduate Research Assistant

2016 to 2018

- DARPA, NSF, and AFRL-funded research on multidimensional RF systems: approximate-DFT beamforming cores and multidimensional delta-sigma ADC architectures for phased-array receivers.

SYNOPSYS INC.

Colombo, Sri Lanka

Product Verification Intern

2015

- Automated product-verification test cases in Perl across VHDL and SystemVerilog ASIC flows with SpyGlass and Xilinx FPGAs.

EDUCATION

FLORIDA INTERNATIONAL UNIVERSITY, Ph.D., Electrical and Computer Engineering, 2020

Miami, Florida

UNIVERSITY OF RUHUNA, B.Sc. (Hons), Electrical and Information Engineering, 2016

Galle, Sri Lanka

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT, A.S., Business Management, 2015

Galle, Sri Lanka

Affiliations: IEEE, member since 2013; Tau Beta Pi Engineering Honor Society, 2019 to present

SELECTED PUBLICATIONS

- Frequency-Multiplexed Array Digitization for MIMO Receivers: 4-Antennas/ADC at 28 GHz on Xilinx ZCU-1285 RF SoC, IEEE Access, vol. 9, 2021.
- Spacetime Frequency-Multiplexed Digital-RF Array Receivers with Reduced ADC Count, IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 68, 2021.
- Massive-MIMO and Digital mm-Wave Arrays on RF-SoCs using FDM for M-Fold Increase in Antennas per ADC/DAC, 2021 IEEE Space Hardware and Radio Conference (SHaRC), 2021.
- Fast Radix-32 Approximate DFTs for 1024-Beam Digital RF Beamforming, IEEE Access, vol. 8, 2020.
- A Direct-Conversion Digital Beamforming Array Receiver with 800 MHz Channel Bandwidth at 28 GHz using Xilinx RF SoC, IEEE COMCAS, Israel, 2019.
- Multiport ADCs for Microwave Focal Plane Array Dish Receivers, IEEE International Symposium on Circuits and Systems, Italy, 2018.

10+ IEEE publications, an AFRL technical report, and a dissertation; [full list at najatthakram.info](http://najatthakram.info).